Light

Legal, Ethical & Environmental Issues

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AQA GCSE Computer Science — **Ethical, Legal and Environmental Impacts (8525 Paper 2)**. This topic covers how digital technology affects society, including privacy issues, environmental concerns, and the ethical considerations of emerging technologies. Exam questions typically ask you to **explain** or **discuss** the impacts of specific technologies on individuals and society. Extended response questions (6—8 marks) are common and require you to consider multiple perspectives, including benefits and drawbacks. You need to understand **general principles** rather than memorise specific details.

Why this topic matters

Digital technology is transforming every aspect of our lives — how we communicate, travel, work, and even how our bodies function. With these changes come important questions that society must answer:

- **Ethical questions:** Just because we *can* do something with technology, *should* we? Who decides?
- **Legal questions:** How do laws keep up with rapidly changing technology? Who is responsible when autonomous systems cause harm?

- **Environmental questions:** What is the carbon footprint of our digital lives? How do we balance technological progress with sustainability?
- **Privacy questions:** How much of our personal data should companies and governments be able to access? Where is the line between security and surveillance?

The AQA specification requires you to understand these issues across several technology areas. You won't be expected to know specific laws or technical details — instead, you need to think critically about the **general principles** and be able to discuss different viewpoints.

What you must know

This checklist covers the key exam-visible concepts from the AQA specification. You should be able to explain and discuss each point from multiple perspectives.

- **Explain** ethical, legal, and environmental impacts and risks of digital technology on society.
- **Consider** data privacy issues where they arise.
- **Apply** your understanding to these technology areas: cyber security, mobile technologies, wireless networking, cloud storage, hacking, wearable technologies, computer-based implants, autonomous vehicles.
- **Understand** that citizens normally value their privacy and may not like governments or security services having too much access to their data.
- **Understand** that governments and security services argue they cannot keep citizens safe from terrorism and other attacks unless they have access to private data.

> **Specification note.** This topic is assessed through extended response questions that require you to discuss issues from multiple angles. You need to understand **general principles** rather than have detailed knowledge of specific issues. Questions will relate to the technology areas listed above.

Toolkit

The three impact categories

Category	Key question	Examples of concerns
Ethical	Is it right or wrong?	Privacy invasion, job displacement, digital divide, addiction, surveillance
Legal	Is it lawful? Who is liable?	Data protection, liability for autonomous systems, intellectual property, hacking laws
Environmental	What is the impact on the planet?	Energy consumption, e-waste, carbon footprint, resource extraction

Technology areas at a glance

Technology	Ethical issues	Legal issues	Environmental issues
Mobile technologies	Addiction, always-on culture, location tracking	Data protection, tracking consent	E-waste, short product lifecycles, rare earth mining
Wireless networking	Health concerns (radiation), digital divide	Unauthorised access, data interception	Energy for infrastructure, equipment disposal
Cloud storage	Who owns data? Trust in providers	Data jurisdiction, breach liability	Data centre energy use, cooling requirements
Cyber security	Privacy vs security trade-off	Prosecution of hackers, international jurisdiction	Energy for security systems
Hacking	White-hat vs black-hat, whistleblowing	Criminal prosecution, varying laws globally	Minimal direct impact
Wearable technologies	Constant monitoring, body data ownership	Health data protection, insurance implications	Battery disposal, frequent upgrades
Computer-based implants	Body autonomy, hacking risks, identity	Medical regulation, consent, hacking liability	Surgical waste, device disposal
Autonomous vehicles	Moral decisions (trolley problem), job losses	Accident liability, insurance, licensing	Electric vs fossil fuel, manufacturing impact

The privacy debate: two perspectives

Citizens' perspective:

- Privacy is a fundamental right
- Mass surveillance feels intrusive
- Data could be misused or leaked
- Chilling effect on free speech

- Distrust of government overreach

Government/security services' perspective:
- Need access to prevent terrorism
- Cannot protect citizens without data
- Criminals use encryption to hide
- National security requires surveillance
- Oversight mechanisms exist

Memory aid: LEEP for impacts

When discussing any technology, consider:
- **L**egal — What laws apply? Who is liable?
- **E**thical — Is it morally right? Who benefits/suffers?
- **E**nvironmental — What resources are used? What waste is created?
- **P**rivacy — What data is collected? Who has access?

1. Understanding ethical, legal and environmental impacts

1.1 Ethical impacts

Ethics is about what is morally right or wrong. Ethical questions in computing often don't have clear answers — different people may have different views based on their values and circumstances.

Key ethical concerns in digital technology include:
- **Privacy:** How much personal information should companies and governments be able to collect and use? Is it ethical to track someone's location, browsing history, or purchases?
- **Surveillance:** Where is the line between legitimate security monitoring and invasive surveillance? Should employers be able to monitor employees' computer use?
- **Job displacement:** As automation and AI replace human workers, who is responsible for those who lose their jobs? Is it ethical to prioritise efficiency over employment?
- **Digital divide:** Not everyone has equal access to technology. Is it fair that some people are excluded from digital services? Whose responsibility is it to provide access?
- **Addiction:** Technology companies design products to be engaging. Is it ethical to create deliberately addictive features, especially for young people?
- **Autonomy:** When computers make decisions that affect people's lives (loan applications, job screening, medical diagnosis), is this fair? Should humans always have the final say?

1.2 Legal impacts

Legal issues concern what is permitted or prohibited by law, and who is responsible when things go wrong. Technology often advances faster than legislation, creating legal grey areas.

Key legal concerns include:
- **Data protection:** Laws govern how personal data can be collected, stored, and used. Organisations must protect data and obtain consent for its use.
- **Liability:** When an autonomous vehicle causes an accident, who is legally responsible — the owner, the manufacturer, or the software developer? Current laws weren't designed for such scenarios.

- **Jurisdiction:** The internet is global, but laws are national. Which country's laws apply when data crosses borders? How can criminals be prosecuted when they operate from other countries?
- **Intellectual property:** Digital content can be copied perfectly and distributed instantly. How do creators protect their work? How should copyright apply to AI-generated content?
- **Hacking and unauthorised access:** Accessing computer systems without permission is illegal, but definitions of "unauthorised access" can be complex. What about security researchers who find vulnerabilities?

1.3 Environmental impacts

Environmental impacts concern the effects of technology on the natural world. While digital technology can seem "clean" compared to heavy industry, it has significant environmental costs.

Key environmental concerns include:

- **Energy consumption:** Data centres that power cloud services consume enormous amounts of electricity. A single large data centre can use as much electricity as a small town. Much of this energy still comes from fossil fuels.
- **E-waste:** Electronic devices contain toxic materials like lead, mercury, and cadmium. When disposed of improperly, these pollute soil and water. The short lifecycle of many devices (phones replaced every 2—3 years) creates mountains of electronic waste.
- **Resource extraction:** Devices require rare earth minerals that must be mined, often in environmentally destructive ways. Manufacturing processes consume water and energy and produce pollution.
- **Carbon footprint:** The technology industry contributes significantly to global carbon emissions — from manufacturing, to shipping, to powering devices and infrastructure.
- **Positive impacts:** Technology can also help the environment through smart energy grids, remote working (reducing commuting), and monitoring climate change. The question is whether the benefits outweigh the costs.

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THE LIFECYCLE ENVIRONMENTAL IMPACT OF A SMARTPHONE

+-------------+     +---------------+     +-------------+     +------
|   MINING    |---->| MANUFACTURING |---->|    USE      |---->| DISPO
+-------------+     +---------------+     +-------------+     +------
      |                    |                    |                  |
      v                    v                    v                  v
 - Rare earth         - Factory           - Charging          - E-wast
   extraction           energy             energy              not re
 - Habitat             - Water use        - Network           - Toxic
   destruction         - Chemical           infrastructure      mater
 - Worker               processes         - Data centres      - Landf
   conditions                                                   pollu
```

```
ENVIRONMENTAL COSTS OCCUR AT EVERY STAGE
Most phones are replaced every 2-3 years, multiplying these impacts
```

Worked Example 1 — Defining ethical issues

Question: What is meant by an ethical issue in computing? Give one example. [2 marks]

Model answer:
An ethical issue is a question about whether something is morally right or wrong **(1)**. For example, whether it is right for companies to track users' locations without their knowledge **(1)**.

Exam tip: Ethical = moral right/wrong. Give a specific, relevant example that shows you understand the concept.

Worked Example 2 — Environmental impacts of cloud storage

Question: Explain one environmental impact of cloud storage. [2 marks]

Model answer:
Cloud storage requires data centres that consume large amounts of electricity to run and cool the servers **(1)**. Much of this electricity comes from fossil fuels, contributing to carbon emissions and climate change **(1)**.

Exam tip: Environmental questions usually relate to energy use, e-waste, or resource extraction. Always explain the consequence, not just state the impact.

Worked Example 3 — Legal issues with autonomous vehicles

Question: Describe one legal issue relating to autonomous vehicles. [2 marks]

Model answer:
If an autonomous vehicle causes an accident, it is unclear who is legally liable **(1)**. It could be the owner, the manufacturer, or the software developer, and current laws were not designed for this situation **(1)**.

Exam tip: Legal issues often involve liability (who is responsible) or jurisdiction (which laws apply). For new technologies, emphasise that laws haven't caught

up.

Worked Example 4 — Ethical issues with wearable technology

Question: Explain one ethical concern about wearable technology that monitors health data. [2 marks]

Model answer:
There is a concern about who owns the health data collected by wearable devices **(1)**. This data could be shared with insurance companies who might use it to increase premiums or deny coverage, which raises questions about fairness **(1)**.

Exam tip: For wearables and implants, focus on body autonomy, data ownership, and the consequences of that data being shared or misused.

2. Technology areas in detail

The specification identifies eight technology areas where you should be able to discuss ethical, legal, and environmental impacts. You need to understand **general principles** that apply to each area, not memorise specific facts.

2.1 Mobile technologies

Mobile devices (smartphones, tablets) have transformed how we live, work, and communicate. They offer unprecedented convenience but raise significant concerns.

Ethical issues: Mobile devices can track our location constantly, creating detailed profiles of our movements. Social media and apps are designed to be addictive, particularly affecting young people. The "always-on" culture blurs boundaries between work and personal life. There are also concerns about the digital divide — those without smartphones are increasingly excluded from services.

Legal issues: Data protection laws govern how companies can collect and use data from mobile devices. Users must give consent for location tracking, but this is often buried in terms and conditions. Who is liable when apps collect more data than users realise?

Environmental issues: Smartphones have short lifecycles (typically 2—3 years before replacement), creating huge amounts of e-waste. Manufacturing requires rare earth minerals extracted through environmentally damaging mining. The constant demand for new models drives resource consumption.

2.2 Wireless networking

Wireless networks (Wi-Fi, mobile data, Bluetooth) enable connectivity without physical cables, but introduce specific concerns.

Ethical issues: Some people have health concerns about electromagnetic radiation from wireless devices and masts, though scientific evidence suggests these are safe at normal levels. The digital divide is also relevant — not everyone has equal access to wireless connectivity, creating social inequality.

Legal issues: Wireless signals can be intercepted more easily than wired connections. Unauthorised access to wireless networks is illegal, but proving it can be difficult. Data transmitted over public Wi-Fi is particularly vulnerable.
Environmental issues: Wireless infrastructure (masts, routers, access points) requires manufacturing and energy. The equipment must eventually be disposed of. However, wireless can reduce the need for physical cables and the resources they require.

2.3 Cloud storage

Cloud storage allows users to store data on remote servers accessed via the internet, rather than on local devices.
Ethical issues: Who truly owns data stored in the cloud? Users may assume they control their data, but service providers have access and could potentially use it. There are questions of trust — can we trust companies to protect our data and not misuse it?
Legal issues: Data jurisdiction is complex — if your data is stored on servers in another country, which country's laws apply? If there is a data breach, who is liable — the user or the cloud provider? Service agreements often favour providers.
Environmental issues: Cloud storage depends on massive data centres that consume enormous amounts of electricity, both for running servers and for cooling them. These data centres have significant carbon footprints, though some companies are moving to renewable energy.

2.4 Cyber security

Cyber security involves protecting systems and data from attack, but the measures taken raise their own concerns.
Ethical issues: There is a fundamental tension between security and privacy. Stronger security measures often require more surveillance and data collection. Is it ethical to monitor all communications to catch a few criminals? The balance between individual privacy and collective security is hotly debated.
Legal issues: Cyber criminals often operate across borders, making prosecution difficult. Different countries have different laws and levels of cooperation. Should companies be legally required to maintain certain security standards? Who is liable when security fails?
Environmental issues: Security systems require computing resources and energy. However, the environmental impact is relatively minor compared to other concerns in this area.

2.5 Hacking (unauthorised access)

Hacking means gaining unauthorised access to a computer system. While often associated with criminals, hacking can have different motivations.
Ethical issues: Is hacking ever justified? "White-hat" hackers find vulnerabilities to help organisations fix them. "Hacktivists" claim to hack for political or social causes. Whistleblowers may leak information they believe the public should know. These raise difficult ethical questions about when breaking the law might be morally justified.
Legal issues: Unauthorised access to computer systems is illegal in most countries, but laws vary. Sentences can range from fines to imprisonment. Prosecution is complicated when hackers operate from different countries. There is debate about whether security researchers who find vulnerabilities should be prosecuted or rewarded.
Environmental issues: Hacking itself has minimal direct environmental impact, though the effects of successful attacks (such as disrupting industrial systems) could have environmental consequences.

2.6 Wearable technologies

Wearable devices (smartwatches, fitness trackers, smart glasses) are worn on the body and often collect health and activity data.
Ethical issues: Wearables enable constant monitoring of our bodies and activities. Who owns this intimate data? Could employers use it to monitor workers? Could insurance companies demand access and adjust premiums based on health data? There are concerns about consent — do users truly understand what data is being collected?
Legal issues: Health data is particularly sensitive and subject to strict data protection rules. However, the line between "health data" and "activity data" can be blurred. If wearables are used in healthcare, they may need to meet medical device regulations.
Environmental issues: Wearables have short lifecycles and contain batteries that require careful disposal. Like smartphones, they require rare earth minerals. The fashion for frequent upgrades increases e-waste.

2.7 Computer-based implants

Computer-based implants are devices placed inside the human body, such as pacemakers, cochlear implants, insulin pumps, or experimental brain-computer interfaces.
Ethical issues: Implants raise profound questions about body autonomy and identity. If a computer is part of your body, who controls it? What happens if the manufacturer goes out of business or stops supporting it? There are concerns about the potential for hacking — what if someone could hack into your pacemaker or brain implant? These technologies also raise questions about human enhancement and what it means to be human.
Legal issues: Medical implants must meet strict regulations, but technology moves faster than regulation. Who is liable if an implant malfunctions? If an implant is hacked, is the hacker liable for assault as well as computer crime? There are also questions about consent — can patients truly give informed consent for technologies they don't fully understand?
Environmental issues: Surgical procedures create medical waste. Implants must eventually be removed or disposed of, and may contain materials that are difficult to recycle. However, the number of implants is relatively small compared to consumer electronics.

2.8 Autonomous vehicles

Autonomous (self-driving) vehicles use sensors, AI, and computing to navigate without human control. They range from cars with driver assistance to fully autonomous vehicles.
Ethical issues: Autonomous vehicles must make split-second decisions that could result in harm. If a crash is unavoidable, should the vehicle prioritise its passengers or pedestrians? (This is often called the "trolley problem.") There are also concerns about job losses for professional drivers (taxi, lorry, bus drivers). Who is responsible for ensuring the AI is fair and doesn't discriminate?
Legal issues: Liability is a major concern. When an autonomous vehicle causes an accident, who is legally responsible? The "driver" who wasn't driving? The manufacturer? The software company? Current insurance and traffic laws weren't designed for vehicles with no human driver. How should these vehicles be licensed and tested?
Environmental issues: Many autonomous vehicles are electric, which could reduce emissions compared to petrol/diesel vehicles. However, manufacturing electric vehicles has significant environmental costs (batteries require lithium mining). The sensors and computing hardware also have manufacturing and disposal impacts. Overall, the

environmental impact depends on whether autonomous vehicles replace private car ownership with shared vehicles, which could reduce the total number of vehicles needed.

Worked Example 5 — Mobile technology impacts

Question: Discuss the ethical and environmental impacts of mobile phone technology. [4 marks]

Model answer:
Ethical: Mobile phones can track users' locations, raising privacy concerns about surveillance **(1)**. Apps and social media are designed to be addictive, particularly affecting young people's mental health **(1)**.
Environmental: Smartphones have short lifecycles, creating large amounts of e-waste containing toxic materials **(1)**. Manufacturing requires mining rare earth minerals, which causes habitat destruction and pollution **(1)**.

Exam tip: When asked to "discuss," include multiple points and explain their significance. Balance ethical and environmental as the question requests.

Worked Example 6 — Autonomous vehicle liability

Question: Explain why liability is a legal concern for autonomous vehicles. [3 marks]

Model answer:
If an autonomous vehicle causes an accident, it is unclear who is legally liable **(1)**. Possible responsible parties include the owner, the vehicle manufacturer, or the software developer **(1)**. Current laws were designed for human drivers and don't adequately address vehicles that drive themselves **(1)**.

Exam tip: For autonomous vehicles, liability is the key legal issue. Name the possible parties and explain why it's complicated.

Worked Example 7 — Computer implant concerns

Question: Describe one ethical and one security concern about computer-based implants. [4 marks]

Model answer:
Ethical: Computer implants raise questions about body autonomy — who controls a device that is inside your body **(1)**? If the manufacturer stops supporting it, the person may have no choice about what happens to part of their body **(1)**.

Security: Implants could potentially be hacked by malicious actors **(1)**. This could have serious consequences if the implant controls vital functions like heart rhythm or insulin levels **(1)**.

Exam tip: For implants, body autonomy and hacking risks are the key concerns. Explain the consequences, not just the risk.

Worked Example 8 – Cloud storage privacy

Question: Explain one privacy concern about storing data in the cloud. [2 marks]

Model answer:
When data is stored in the cloud, the cloud provider has access to that data **(1)**. Users may not know who can view their data or whether it might be shared with third parties or governments **(1)**.

Exam tip: Cloud privacy concerns focus on loss of control — your data is on someone else's servers, and you don't fully control who can access it.

3. The privacy debate

The AQA specification specifically requires you to understand the tension between individual privacy and collective security. This is one of the most important debates in digital society.

3.1 The citizen's perspective

Ordinary citizens normally value their privacy and may be uncomfortable with governments or security services having extensive access to their personal data. Key arguments from this perspective include:

- **Privacy as a right:** Privacy is considered a fundamental human right in many legal systems. People should be free from constant surveillance.
- **Chilling effect:** If people know they are being watched, they may censor themselves. This can harm free speech and political participation, even for those with nothing to hide.
- **Risk of misuse:** Data collected for security purposes could be misused. Governments change, and data collected under one government could be abused by future governments.
- **Data breaches:** Large databases of personal information are attractive targets for hackers. The more data that is collected, the more there is to steal.
- **Disproportionate surveillance:** Mass surveillance treats everyone as a potential suspect. Many argue that targeted surveillance of actual suspects is more proportionate and effective.
- **Function creep:** Powers granted for one purpose (fighting terrorism) tend to expand to other purposes (investigating minor crimes, monitoring protesters).

3.2 The government/security services' perspective

Governments and security services argue that they cannot keep citizens safe from terrorism and other serious threats unless they have access to private data. Key arguments from this perspective include:

- **Preventing terrorism:** Modern terrorists use encrypted communications and the internet to plan attacks. Security services argue they need access to communications to prevent attacks before they happen.
- **Serious crime:** Criminals also use technology. Access to data helps solve serious crimes and bring perpetrators to justice.
- **Nothing to hide:** Some argue that law-abiding citizens have nothing to fear from surveillance — only those with something to hide need be concerned.
- **Oversight exists:** In democracies, surveillance is subject to legal oversight. Warrants are required. Agencies are accountable to elected representatives.
- **Balancing act:** Security services argue they are balancing privacy against the need to protect life. Some loss of privacy is the price of security.
- **Criminals going dark:** If law enforcement cannot access encrypted communications, criminals can communicate with complete impunity. This is sometimes called the "going dark" problem.

3.3 Finding a balance

There is no easy answer to the privacy debate. Most people accept that some surveillance is necessary for security, but disagree about how much is proportionate. Key considerations include:

- **Proportionality:** Are the surveillance measures proportionate to the threat? Mass surveillance of everyone may be disproportionate compared to targeted surveillance of suspects.
- **Oversight:** Who watches the watchers? Strong independent oversight can help ensure powers are not abused.
- **Transparency:** People should know, in general terms, what surveillance powers exist and how they are used.
- **Minimisation:** Collect only the data that is necessary, retain it only as long as needed, and delete it when no longer required.
- **Purpose limitation:** Data collected for security should only be used for security, not for other purposes.

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THE PRIVACY VS SECURITY SPECTRUM

+-------------------------------------------------------------------+
|                                                                   |
|   MAXIMUM PRIVACY                           MAXIMUM SECURITY      |
|   <---------------------------------------------->                |
|                                                                   |
|   - No surveillance                     - Mass surveillance       |
|   - Complete encryption                 - No encryption allowed   |
|   - No data collection                  - Collect all data        |
|   - Total anonymity                     - Identity verified always |
```

```
|                                                                    |
|   Risks:                                Risks:                     |
|   - Criminals escape                    - Privacy violations       |
|   - Terrorism harder                    - Chilling effect          |
|     to prevent                          - Abuse of power           |
|                                                                    |
+--------------------------------------------------------------------+

MOST PEOPLE WANT SOMETHING IN BETWEEN
The debate is about WHERE to draw the line
```

Worked Example 9 — Privacy concerns of citizens

Question: Explain why ordinary citizens might be concerned about government access to their data. [3 marks]

Model answer:
Citizens value privacy as a fundamental right and may not want to be under constant surveillance **(1)**. There is concern that data collected could be misused or leaked in a data breach **(1)**. Mass surveillance may create a "chilling effect" where people censor themselves because they know they are being watched **(1)**.

Exam tip: This is a specification-required viewpoint. Key points: privacy as a right, risk of misuse, chilling effect on free speech.

Worked Example 10 — Government perspective on data access

Question: Explain why governments argue they need access to private data. [3 marks]

Model answer:
Governments argue they cannot protect citizens from terrorism without access to communications data to identify and monitor potential threats **(1)**. Criminals use encrypted communications, making it harder for law enforcement to investigate serious crimes **(1)**. They argue that surveillance powers are subject to legal oversight and are proportionate to the threats faced **(1)**.

Exam tip: This is a specification-required viewpoint. Key points: preventing terrorism, investigating crime, oversight exists.

Worked Example 11 – Discussing both perspectives

Question: Discuss whether governments should have access to encrypted communications. [6 marks]

Model answer:
Arguments for access: Governments argue that terrorists and criminals use encrypted messaging to plan attacks and crimes. Without access, law enforcement cannot investigate or prevent serious harm. They suggest that with proper legal oversight, access can be granted proportionately.
Arguments against access: Citizens value privacy as a fundamental right. Creating "backdoors" in encryption would weaken security for everyone, including against foreign governments and criminals. Mass access to communications could have a chilling effect on free speech. There is also the risk that access powers could be misused or expanded beyond their original purpose.
Conclusion: The debate involves balancing legitimate security needs against fundamental rights to privacy. Most agree that some targeted access may be justified with proper oversight, but mass surveillance of all communications would be disproportionate.

Exam tip: For extended response questions, present both sides fairly, use clear paragraphs, and offer a balanced conclusion. Don't just list points — explain and develop them.

Worked Example 12 – Extended response on technology impacts

Question: Discuss the ethical, legal, and environmental impacts of wearable health-monitoring technology. [8 marks]

Model answer:
Ethical impacts: Wearable devices collect intimate data about users' bodies and activities, raising questions about who owns this data. There is concern that employers or insurance companies could demand access to this data, using it to make decisions about employment or premiums, which could be seen as unfair. Users may not fully understand what data is collected or how it is used, raising questions about informed consent. On the positive side, wearables can help people monitor their health and encourage healthier lifestyles.

Legal impacts: Health data is particularly sensitive and subject to data protection laws. However, the boundary between "health data" and "fitness data" is unclear. If companies share this data without proper consent, they could face legal action. There are also questions about whether wearables used for health purposes should be regulated as medical devices, which would require more rigorous testing and approval.
Environmental impacts: Wearables have short product lifecycles as technology improves and fashion changes. This creates electronic waste containing batteries and potentially toxic materials that must be disposed of carefully. Manufacturing requires rare earth minerals that are extracted through environmentally damaging mining processes. The constant demand for new devices increases resource consumption and carbon emissions from manufacturing and shipping.
Conclusion: Wearable technology offers genuine health benefits but raises significant concerns about data privacy, legal protection, and environmental sustainability. Addressing these concerns requires clear regulation about data ownership and use, and encouraging longer device lifespans and better recycling.

Exam tip: For 8-mark extended responses, structure your answer with clear paragraphs for each impact type. Include both positives and negatives where relevant, and finish with a conclusion that pulls the discussion together.

Worked examples summary

The worked examples throughout this topic cover the main exam question styles you'll encounter. Here's a summary of what each example demonstrates:

Example	Topic	Marks	Skill tested
1	Defining ethical issues	2	Definition with example
2	Environmental impacts of cloud	2	Explain with consequence
3	Legal issues with autonomous vehicles	2	Describe legal concern
4	Ethical issues with wearables	2	Explain with consequence
5	Mobile technology impacts	4	Discuss multiple impacts
6	Autonomous vehicle liability	3	Explain legal complexity
7	Computer implant concerns	4	Describe ethical and security
8	Cloud storage privacy	2	Explain privacy concern

Example	Topic	Marks	Skill tested
9	Citizens' privacy concerns	3	Explain perspective
10	Government data access argument	3	Explain perspective
11	Encryption debate	6	Extended response, both sides
12	Wearable technology impacts	8	Full extended response

Quick reference

Key definitions

Term	Definition
Ethical issue	A question about what is morally right or wrong.
Legal issue	A question about what is lawful or who is liable/responsible.
Environmental issue	A concern about impact on the natural world (energy, waste, resources).
Privacy	The right to control personal information and be free from surveillance.
Liability	Legal responsibility for something, especially when it goes wrong.
E-waste	Discarded electronic devices, often containing toxic materials.

Technology areas summary

Technology	Key ethical concern	Key legal concern	Key environmental concern
Mobile tech	Addiction, tracking	Data protection	E-waste, short lifecycles
Wireless	Digital divide	Interception, access	Infrastructure energy
Cloud storage	Data ownership, trust	Jurisdiction, breaches	Data centre energy
Cyber security	Privacy vs security	International jurisdiction	Energy use
Hacking	White-hat vs black-hat	Prosecution, varying laws	Minimal

Technology	Key ethical concern	Key legal concern	Key environmental concern
Wearables	Constant monitoring	Health data protection	Battery disposal
Implants	Body autonomy, hacking	Medical regulation	Device disposal
Autonomous vehicles	Moral decisions, job losses	Accident liability	Electric vs fossil, manufacturing

The privacy debate summary

Citizens argue:

- Privacy is a fundamental right
- Mass surveillance is intrusive
- Data could be misused
- Chilling effect on free speech

Governments argue:

- Need access to prevent terrorism
- Criminals use encryption
- Oversight mechanisms exist
- Security requires some surveillance

Typical pitfalls

Pitfall 1 — Only giving one side.
Wrong: Just listing advantages or just disadvantages.
Right: Extended response questions require you to consider multiple perspectives and present a balanced discussion.

Pitfall 2 — Being too vague.
Wrong: "There are privacy concerns."
Right: Explain specifically what the privacy concern is and why it matters — "Companies can track users' locations, which could be misused or leaked."

Pitfall 3 — Confusing ethical and legal.
Wrong: Mixing up what is morally wrong with what is illegal.
Right: Ethical = is it right/wrong? Legal = is it lawful/who is liable? Something can be legal but unethical, or illegal but ethical.

Pitfall 4 — Forgetting environmental impacts.
Wrong: Only discussing ethical and legal issues.

Right: Remember energy consumption, e-waste, and resource extraction — these apply to almost all technologies.

Pitfall 5 — Not applying to context.
Wrong: Generic statements that could apply to any technology.
Right: Make your points specific to the technology in the question — autonomous vehicles have different issues than wearables.

Pitfall 6 — Ignoring the privacy debate.
Wrong: Not mentioning the tension between privacy and security.
Right: The spec explicitly requires you to understand both citizens' and governments' perspectives on privacy.

Pitfall 7 — Lists instead of discussion.
Wrong: Bullet point lists of facts.
Right: Extended responses should use paragraphs, explain and develop points, and show reasoning.

Pitfall 8 — Memorising specific details.
Wrong: Trying to remember specific laws or case studies.
Right: The spec says you need "general principles" not "detailed knowledge on specific issues."

Pitfall 9 — Forgetting liability for autonomous vehicles.
Wrong: Only discussing the "trolley problem" for autonomous vehicles.
Right: Liability (who is responsible when things go wrong) is the key legal issue for autonomous vehicles.

Pitfall 10 — Missing the "so what?".
Wrong: "Wearables collect health data."
Right: Explain the consequence: "This data could be shared with insurers who might increase premiums or deny coverage."

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